Tristin Xie

(626) 268-3247 | tristinxie@gmail.com | tristinxie.com | tristinxie | tristinxie

Education

University of California San Diego

M.S. in Computer Science — *Sep. 2023 - Present*

- **Relevant Coursework:** Robotics · Computer Vision · Machine Learning · Natural Language Processing · Recommender Systems · Deep Learning

University of California San Diego — *CS GPA: 3.722 / 4.0*

Double Major: B.S. in Computer Science · B.S. in Applied Mathematics · Minor in Business — *Sep. 2019 - Sep. 2023*

- **Tau Beta Pi:** Oldest engineering honor society with invitations only sent to the **top 1/8th of the Junior class and top 1/5th of the Senior class**
- **Relevant Coursework:** Advanced Data Structures and Algorithms · Software Engineering · Operating Systems · Computer Architecture

Work Experience

Advanced Robotics and Controls Lab (ARClab) at UC San Diego — *San Diego, CA*

Research Assistant — *Mar. 2023 - Present*

- Published CtRNet-X: Camera-to-Robot Pose Estimation in Real-world Conditions Using a Single Camera to ICRA 2025 as second author
- Created a data collection pipeline for the **Franka Emika Panda** robot and **Microsoft Kinect** through **Robot Operating System (ROS)**
- Wrote a **particle filter** with **COLMAP**, **ORB-SLAM3**, **DPVO** motion models in **PyTorch** for robust camera-to-robot pose estimation

Amazon Web Services — *Seattle, WA*

Software Development Engineer Intern — *Jun. 2023 - Sep. 2023*

- Overhauled **autocomplete** from a static frontend approach to a dynamic backend approach reflecting real data in **Elasticsearch**
- Tested implementation by writing new **unit, integration, and canary tests** and created **AWS CloudWatch alarms, metrics, and dashboards**
- **Improved search results by 28%** for exact keyword match queries of a major category, diagnosed by writing a **multithreaded Python** script

Software Development Engineer Intern — *Jun. 2022 - Sep. 2022*

- Implemented and tested an **AWS DynamoDB** schema and **AWS Lambda** API using **Go** to support Word-like notes for **10,000+ AWS partners**
- Defined cloud infrastructure using **AWS Cloud Development Kit** with **TypeScript** and deployed **CloudFormation** stacks for my application

UC San Diego Computer Science and Engineering — *San Diego, CA*

Instructional Assistant — *Jul. 2021 - Mar. 2023*

- Tutored preparatory and lower division courses on **algorithms, recursion, state machines, Pygame, Python, and image manipulation**
- Tutored upper division course with assigned **C++** programming assignments and conceptual problems for a combined total of **100+ hours**
- Tested **30+** advanced data structures assignments including **self-balancing binary search trees, tries, hash maps, bloom filters, and more**

Leadership

Association for Computing Machinery at UC San Diego — *San Diego, CA*

Vice President Projects — *May 2022 - May 2023*

- Oversaw the processing of **250+ applicants** and hosted a project showcase event with **90+ attendance** for the quarterly ACM Projects program

Vice President Finance — *May 2021 - May 2022*

- Directed the finance team to create financial documents to **obtain funding** for **200+ events** from school departments and student government
- Managed and budgeted **thousands of dollars** of department funds, sponsorship money, and other miscellaneous University allocated funds

Projects

ACM Teammate Finder: Web App To Find Teammates Easily | GitHub — *Oct. 2020 - Jan. 2021*

- Implemented **Material Design** webpages with **React, HTML/CSS, and React Material Web Components**
- Developed **CRUD functions** for back-end using **Express** to fetch and post updates to the **Firestore** database

EnviroSense: Internet of Things Pollution Sensor | Devpost | GitHub — *Oct. 2019*

- **SDHacks WINNER (3rd):** *Best IoT that Incorporates Multiple Nodes (team of 4)*

PictureTranslate: Real-time Computer Vision Object Translator | GitHub — *Aug. 2019*

- Trained **Haar Cascades** on everyday objects and augmented video feed with translation of detected object using the **Google Translate API**

Skills

Languages	Python · C/C++ · Go · JavaScript/TypeScript · Haskell · HTML/CSS · Sass · ARM Assembly · Bash · Java · LaTeX
Frameworks and Tools	AWS · Git · React · Node.js · Express.js · NumPy · Tailwind CSS · Firebase · OpenCV · Jest · GDB · Netlify · PyTorch
Software	Vim · Visual Studio Code · Visual Studio · IntelliJ IDEA Ultimate · Adobe Illustrator